



École d'ingénieurs Polytech Orléans

GPSE - Génie Physique et Systèmes Embarqués

Rapport d'avancement équipe 1

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Introduction

The project is the French Robotics Cup 2027; the goal is to make a robot to participate in the competition. The project was proposed by Robotek Orléans. This first report concentrates on the start of the project and project management.

1 French Robotics Cup 2027

The French Robotics Cup (CFR) is an annual competition that takes place in May. For 2027, the competition starts on May 5th and ends on May 9th. The competition is organized by Eurobot. Each year, the competition has a different theme; this year's theme is Kaamelott, and the different tasks are inspired by it. The competition consists of matches between two different teams, with a match duration of 100 seconds. The duel takes place in an area designed by Eurobot, which is available in the first appendix. To participate in the competition, the robot needs to pass the homologation phase. The goal of this phase is to ensure that the robot is not dangerous for the team or the opponents. The rulebook gives a different number of points depending on the tasks completed by the robot.

This year, the first task is to make a castle with towers and walls to protect the Grail. The second task is to attack the opponent's castle with the balls provided by the organizers. For more details, please see the second appendix.

2 User case table

Scenario	**Step**
Start of the match	1) Put the robot in the battle area
	2) Put the balls inside the robot
	3) Start the robot
Creation of the walls	1) Move forward to the location of the crates
	2) Collect the crates
	3) Move around until the location of the walls
	4) Deposit the crates
Creating Towers	1) Move forward to the location of the crates
	2) Return a crate
	3) Move to the location of the towers
	4) Deposit the cash register
Ball shooting	1) Move forward to the location of the crates
	2) Throw the ball
Reload	1) Stop the robot
	2) Take the power cell
	3) Connect the power loader
	4) Activate the generator
	5) Turn on the power loader
Reusability	1) Put away the robot
	2) Dismantle the robot
	3) Take the different components

3 Functional analysis

3.1 Operational entities

The robot encounters different entities like the other robots, the blocks, or the secondary robot named PAMI. The robot needs to interact with or avoid the different obstacles.

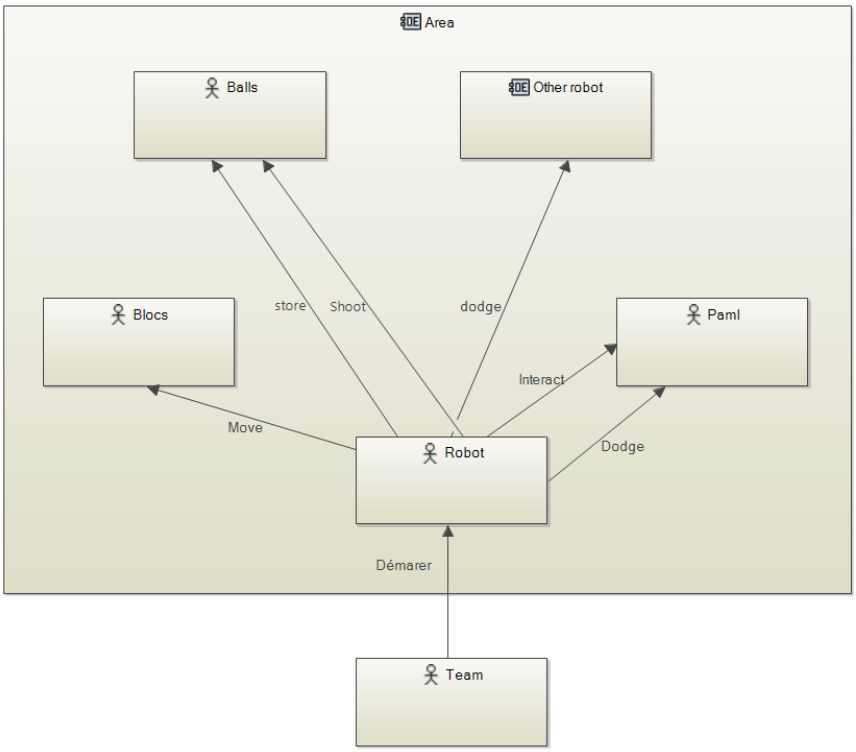


Figure 1: OAB diagram

3.2 Mission and capabilities

The robot has different capabilities, such as being autonomous or avoiding other robots.

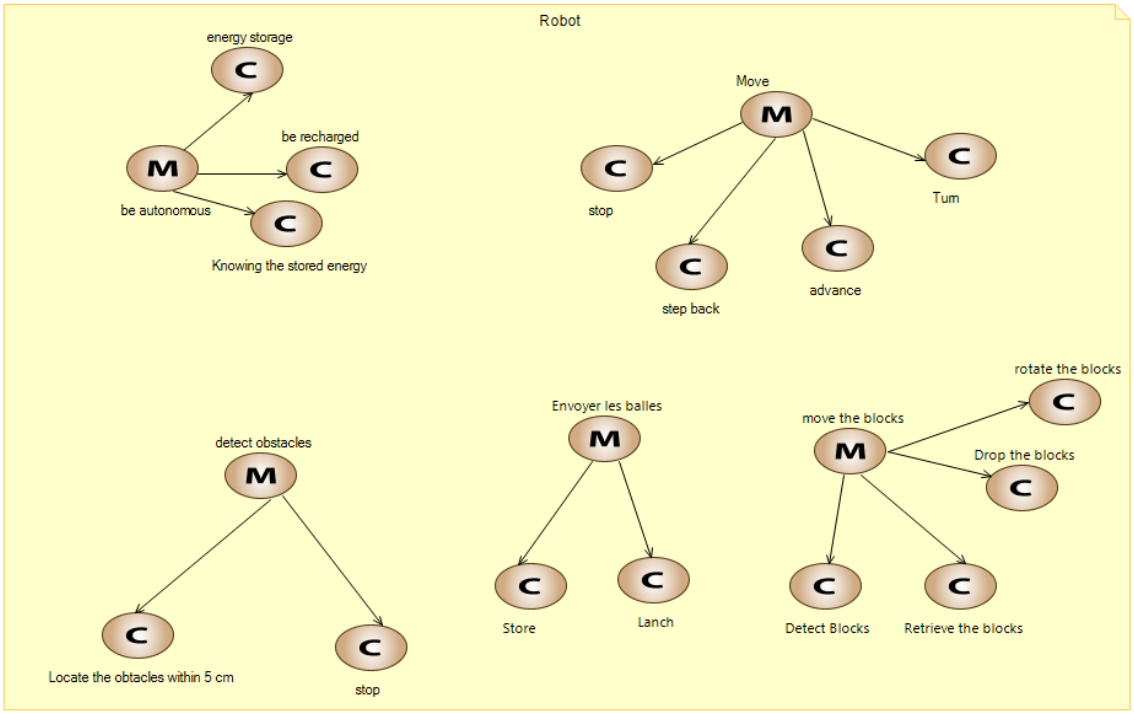


Figure 2: MCB diagram

3.3 Internal structure

The robot is made of different parts like the base for the motors and movement, or the obstacle avoidance system.

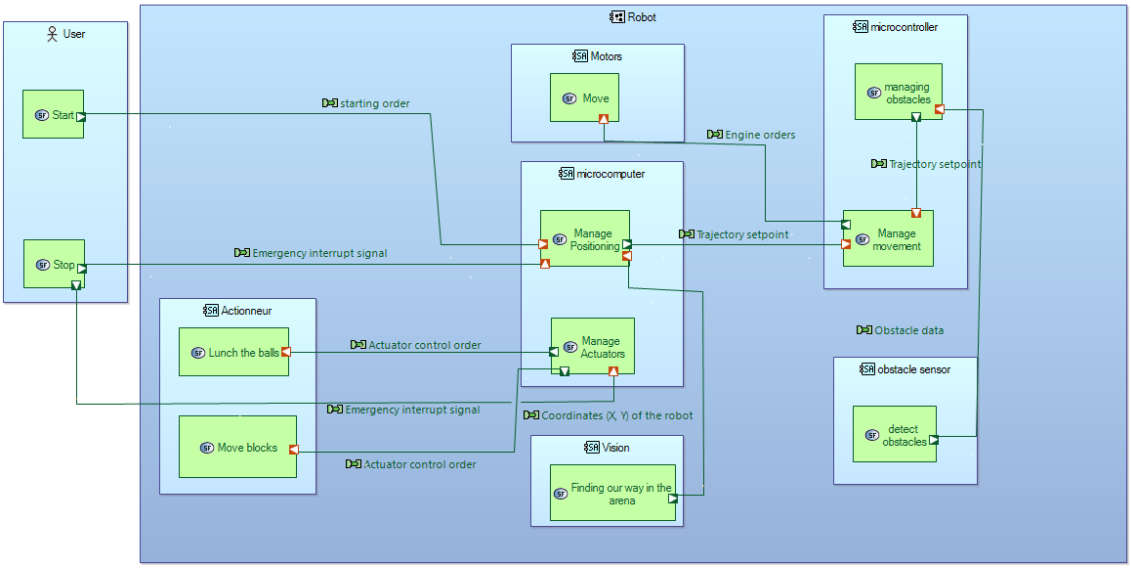


Figure 3: SAB Diagram

3.4 Listing of functions

The different diagrams allowed us to create this list of functions:

ID	Name	Description	Component	Tech.	Input	Output	Criterion	Value	Tol.	Prio.
FP1	Move	Move autonomously	Microcontroller	C++	Command	PWM	-	-	-	P1
			4 DC motors	Electro.	PWM	Rotation	Speed (rad/s)	6.25	± 0.5	
			4 Wheels	Mech.	Rotation	Trans.	Speed (m/s)	0.25	± 0.1	
			H-bridge	Power	PWM+12V	6V voltage	-	-	-	
FP2	Obstacle	Detect obstacle	Microcontroller	C++	Digital	Stop cmd	Reaction time	10 ms	± 2 ms	P1
			Ultrasonic sen.	Electro.	Wave	Digital	Distance (cm)	5 cm	± 2 cm	
FP3	Power	Self-powered	Battery	Power	Energy	12V/5V	Operating time	Min 200s	-	P1
			-	-	-	-	Capacity (mAh)	Min 3000	-	
FP4	Locate	Read QR/Tags	Microcontroller	Python	Read cmd	-	-	-	-	P1
			Camera	Optical	-	Pixels	Reading dist.	100 cm	± 10	
FP5	Blocks	Move & rotate	Microcontroller	Python	/	PWM	-	-	-	P2
			Servomotors	Electro.	PWM	Rotation	Rotation ($^\circ$)	180°	$\pm 5^\circ$	
			Pneumatic mod.	Pneum.	Electric	Linear	Pressure (bar)	-	-	
FP6	Shoot	Shoot balls	Microcontroller	Python	Shoot cmd	PWM	-	-	-	P2
			Motor	Electro.	PWM	Rotation	Speed (rad/s)	150	± 10	
FP7	Grail	Elevate Grail	Microcontroller	Python	Lift cmd	PWM	-	-	-	P2
			Servomotors	Electro.	PWM	Rotation	Height (mm)	350	± 50	
FC1	E-Stop	Emergency stop	Push button	Interrup.	Physical	Power cut	Stop time	Max 500ms	-	P1
FC2	Dimensions	Respect size	Undeployed dim.	/	-	-	Perimeter (mm)	Max 1200	-	P1
			Deployed dim.	/	-	-	Perimeter (mm)	Max 1400	-	

Table 1: Listing of functions and technical specifications

4 Planning

To establish our schedule, we used GanttProject, which proved very helpful for planning the different project steps and distributing tasks.

To distribute tasks efficiently, we first identified what each member preferred not to do. Afterward, we assigned specific domains of activity based on everyone's preferences:

- **Maxime Fabre** focused on the robot's vision system for obstacle detection and localization.
- **Rayan Guedouir** managed the power system, ensuring monitoring of the energy stored inside the robot.
- **Thibaut Martin** worked on the mechanical structure and movement systems.
- **Mickaël Borges** focused on the actuators.

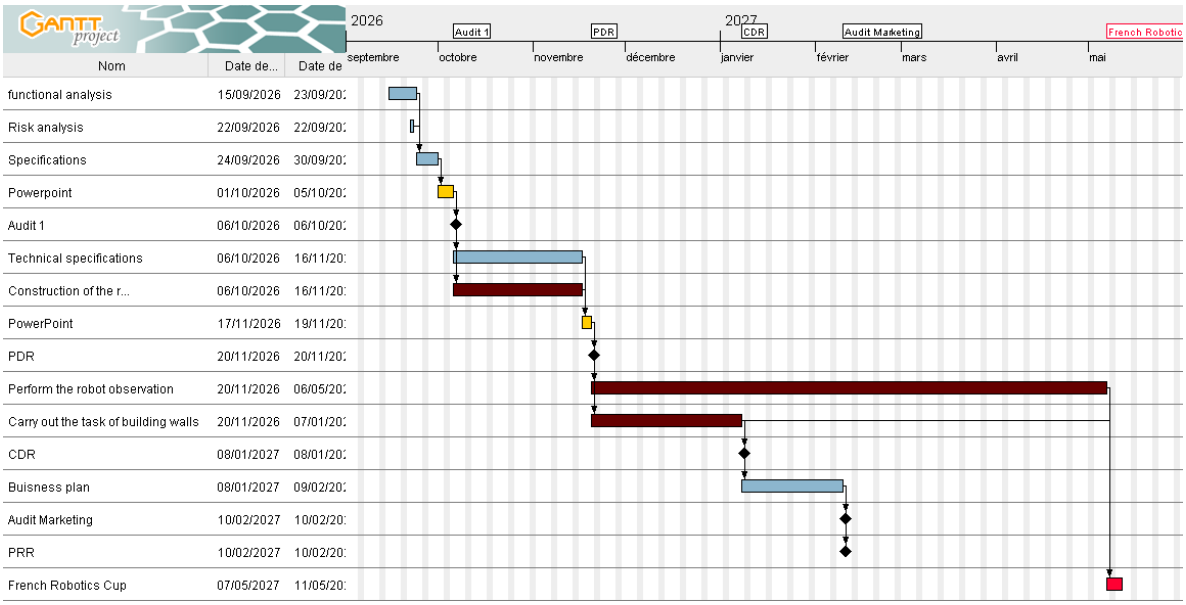


Figure 4: Gantt Project Schedule

5 Personal feedback

5.1 Maxime Fabre

5.2 Thibaut Martin

5.3 Rayan Guedouir

5.4 Mickaël Borges